

ORIGINAL ARTICLE

HTD1801 in Combination with Metformin for Type 2 Diabetes

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Abstract

BACKGROUND HTD1801 is an anti-inflammatory metabolic modulator being developed to treat type 2 diabetes (T2D). The safety and efficacy of HTD1801 as an add-on to metformin among patients with T2D with inadequate glycemic control despite stable metformin use have not been well evaluated.

METHODS This phase 3 double-blind trial included patients with T2D, glycated hemoglobin (HbA1c) 7.0% to 10.5%, fasting glucose levels of 250.5 mg/dl or lower, and stable use of metformin for 8 weeks or more. Patients were randomly assigned 2:1 to receive either HTD1801 1000 mg orally twice daily (n=365) or placebo (n=184) in addition to their prior metformin treatment. The primary end point was change in HbA1c from baseline to week 24. Data on adverse events were collected during this 24-week time frame.

RESULTS The mean HbA1c was 8.6% at baseline and 7.4% at 24 weeks in the HTD1801 group (mean change, −1.2 percentage points; 95% confidence interval [CI], −1.3 to −1.1). The mean HbA1c was 8.5% at baseline and 7.8% at 24 weeks in the placebo group (mean change, −0.7 percentage point; 95% CI, −0.8 to −0.6). The adjusted least-squares mean difference between groups for the change in HbA1c from baseline to week 24 was −0.5 (95% CI, −0.7 to −0.4; P<0.0001). Diarrhea was the most common adverse event (HTD1801, 23.8%; placebo, 1.1%), with three cases in the HTD1801 group graded as severe, and all the other cases being mild to moderate in severity. No patients experienced severe hypoglycemia.

CONCLUSIONS After 24 weeks, HbA1c reduction was significantly greater among patients with T2D taking metformin and randomly assigned to receive HTD1801 versus placebo. A higher proportion of patients in the HTD1801 group experienced mild to moderate diarrhea compared with the placebo group. (Funded by Shenzhen HighTide Biopharmaceutical; ClinicalTrials.gov number, [NCT06353347](https://clinicaltrials.gov/ct2/show/study/NCT06353347).)

Introduction

Type 2 diabetes (T2D) is a multifaceted metabolic disorder characterized by insulin resistance and low-grade inflammation and is reportedly the eighth leading cause of death worldwide.^{1,2} Patients with T2D typically present with hyperglycemia

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and also commonly have concomitant risk factors for other metabolic, renal, and cardiovascular disorders that can exacerbate disease.¹ A prior cross-sectional, multicenter observational study reported that patients with concurrent T2D, hypertension, and dyslipidemia were six times more likely to have a history of cardiovascular disease compared with patients with T2D alone.³ Metformin is the most commonly prescribed oral antidiabetic medication worldwide; however, one-third of treated patients do not achieve glycemic control with metformin alone.⁴⁻⁷ Furthermore, metformin does not address the other metabolic, renal, and cardiovascular comorbidities associated with T2D.

HTD1801 is an anti-inflammatory metabolic modulator being investigated as an oral treatment for metabolic diseases. HTD1801 activates adenosine monophosphate-activated protein kinase and inhibits the nucleotide-binding oligomerization domain-like receptor-family pyrin domain containing-3 inflammasome to increase fatty acid oxidation and glucose utilization while reducing insulin resistance and chronic inflammation. HTD1801 has been evaluated in multiple clinical trials for metabolic and non-viral liver diseases.⁸⁻¹⁰ In a phase 2 trial in patients with T2D, HTD1801 resulted in a reduction in glycated hemoglobin (HbA1c) of -0.7 percentage points at 12 weeks (mean at baseline of 8.2%) and improvements in key cardiometabolic parameters and markers of inflammation.¹¹

Based on this prior work, three phase 3 trials evaluating HTD1801 for the treatment of T2D were recently conducted: Efficacy and Safety of HTD1801 in Patients With Type 2 Diabetes (SYMPHONY-1) (monotherapy),¹² Efficacy and Safety of HTD1801 in Patients With Type 2 Diabetes Inadequately Controlled With Metformin (SYMPHONY-2) (add-on to metformin), and Effects of HTD1801 Versus Dapagliflozin in Patients With Type 2 Diabetes Inadequately Controlled With Metformin (HARMONY) (add-on to metformin compared with dapagliflozin).¹³ In the SYMPHONY-1 trial, HTD1801 administered as monotherapy resulted in a least-squares (LS) mean change in HbA1c of -1.3 percentage points compared with -0.6 percentage point with placebo with an LS mean difference between groups of -0.66 ($P<0.0001$) at 24 weeks (primary outcome).¹²

The current article reports the results of SYMPHONY-2, evaluating the efficacy and safety of HTD1801 as an add-on to metformin in patients with T2D and inadequate glycemic control.

Methods

PATIENTS

Patients were screened and recruited at 64 participating hospitals across China. Patients aged 18 years or older with a diagnosis of T2D (per World Health Organization criteria¹⁴) receiving a stable dose of metformin (≥ 1500 mg/d or maximum tolerated dose [≥ 1000 mg/d]) for at least 8 weeks, with an HbA1c between 7.5% and 11.0% and with a fasting plasma glucose (FPG) level of 250.5 mg/dl or lower, were eligible to enter a 4-week run-in period. During the run-in period, patients were required to adhere to standardized dietary and physical activity recommendations, perform glucose self-monitoring, and undergo assessment of adherence to their ongoing metformin regimen.¹⁵ One week before random assignment, eligibility was reassessed by using protocol-defined glycemic thresholds (HbA1c, 7.0%-10.5%; FPG, ≤ 250.5 mg/dl). Only patients who met all re-evaluation criteria were eligible for random assignment. Additional details on eligibility are provided in Section 3.3 in the Supplementary Appendix.

TRIAL DESIGN AND OVERSIGHT

SYMPHONY-2 was a phase 3 trial that enrolled patients with T2D experiencing inadequate glycemic control with metformin monotherapy. The trial included a 4-week single-blind placebo run-in period followed by a 24-week double-blind placebo-controlled treatment period, then a 28-week open-label extension period, and then a final 4-week safety follow-up period evaluation (i.e., total of 56 weeks for the safety evaluation) (Fig. S1).

Randomization was performed with an Interactive Web Response System (Clinflash). A block randomization method was used with baseline HbA1c ($<8.5\%$ or $\geq 8.5\%$) and high-sensitivity C-reactive protein (hs-CRP; ≤ 3 mg/l or >3 mg/l) values as stratification factors. Patients were randomly assigned 2:1 to receive either oral HTD1801 1000 mg or placebo twice daily as add-on therapy to metformin. Treatment was provided as matching capsules. Patients, trial personnel, and the sponsor were blinded to treatment assignment. All patients in the open-label extension received HTD1801 1000 mg twice daily.

Following a baseline assessment, in-person visits occurred at weeks 6, 12, 18, 24, 31, 38, 45, 52, and 56. Blood samples were collected with the patient in the fasting state except for postprandial samples, which were collected following a sponsor-provided standardized meal (Section

3.5 of the Supplementary Appendix provides details on the mixed meal tolerance test). The homeostasis model assessment of insulin resistance was calculated by using standard formulas.^{16,17} According to Chinese T2D guidelines, hypoglycemia was defined as a blood glucose level less than 70.3 mg/dl.¹⁵ FPG levels were monitored at each visit, and rescue therapy was initiated based on prespecified criteria described in Section 3.4 of the Supplementary Appendix.

The protocol (provided with the full text of this article at evidence.nejm.org) was approved by the ethics committees of all participating centers, and all participants provided written informed consent. This trial was conducted in accordance with the Declaration of Helsinki and regulations consistent with the International Council for Harmonisation Guidelines for Good Clinical Practice, and it followed Consolidated Standards of Reporting Trials guidelines.¹⁸ The authors vouch for the accuracy and completeness of the data and analyses reported and for the fidelity of this report to the protocol. Additional information about trial oversight is provided in Section 3.1 of the Supplementary Appendix.

END POINTS

The primary end point was the difference in the change in HbA1c from baseline to week 24 between the HTD1801 and placebo groups. Key prespecified secondary end points included the between-group difference in the change from baseline to 24 weeks for the following measures: FPG, LDL cholesterol (LDL-C), non-HDL cholesterol (non-HDL-C), gamma-glutamyl transferase (GGT), and hs-CRP. The proportion of patients achieving an HbA1c value less than 7.0% by 24 weeks was an additional key secondary end point. The change in all these measures from baseline to week 52 were also included as secondary end points. A complete list of efficacy end points is provided in Section 3.2 of the Supplementary Appendix.

ADVERSE EVENTS

Adverse events were monitored from baseline through week 56. Section 8 of the protocol details the collection, definitions, and reporting of adverse events. At each scheduled visit, participants were asked in an open-ended fashion whether they experienced any new or ongoing medical events. Adverse events were also ascertained from physical examinations, vital sign measurements, laboratory testing, and review of concomitant medications. Adverse events were coded according to system organ class and preferred term using the Medical Dictionary for Regulatory Activities

Chinese version 28.0.¹⁹ Adverse events were graded by using the National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0.²⁰ Serious adverse events were reported in accordance with applicable regulatory requirements in China. Hypoglycemic events were defined in accordance with guidelines for prevention and treatment of T2D in China (2020) as follows: Grade 1, blood glucose less than 70.2 mg/dl but 54 mg/dl or higher mg/dl; Grade 2, blood glucose less than 54 mg/dl; and Grade 3, a serious event with no specific blood glucose threshold, accompanied by changes in consciousness and/or body, and requiring help from others.¹⁵

STATISTICAL ANALYSIS

Sample size was calculated based on a one-sided Student's t-test (significance level of 0.025) and the following assumptions for HTD1801 compared with placebo: mean difference in HbA1c from baseline to week 24 of 0.5 percentage point, pooled standard deviation (SD) of the change in HbA1c from baseline to week 24 of 1.3 percentage points, and 2:1 randomization. With these assumptions, an estimated total sample size of 324 patients would provide 90% power to detect a significant difference in change in HbA1c between HTD1801 and placebo. Considering a 20% dropout rate, and to meet regulatory guidelines on required safety exposure, the planned sample size was 555 patients.

Statistical testing was two-sided and was performed at the 0.05 alpha level. All statistical analyses in this trial were performed by using SAS version 9.4 or later (SAS Institute). The full analysis set included all patients who received at least one dose of treatment, and the safety analysis set included all patients who received at least one dose of treatment and had at least one postdose safety assessment. Efficacy measures were assessed by using a treatment policy strategy that included all observations after intercurrent events of rescue therapy or premature treatment discontinuation.

In the primary end point analysis, missing data were assumed to be missing at random and were analyzed by using a mixed-effects model for repeated measures with the change in HbA1c from baseline as the dependent variable. Independent variables included treatment group, measurement time point, the interaction between treatment group and measurement timepoint, and the randomization stratification factors. Stratification factors included baseline hs-CRP category (≤ 3 mg/l vs. > 3 mg/l) and HbA1c as a continuous covariate. Superiority was judged based on the model-adjusted LS mean difference between groups and the two-sided 95% confidence interval (CI).

To evaluate the robustness of the primary analysis to different assumptions regarding missing data and analytical approach, sensitivity analyses included an analysis of covariance using the last-observation-carried-forward and mixed-effects model for repeated measures using multiple imputation. Details regarding the sensitivity analyses are provided in Section 3.6.1 of the Supplementary Appendix.

For key secondary end points, type I error was controlled for by using a prespecified sequential gatekeeping strategy (Section 3, Statistical Analysis Plan). Testing proceeded sequentially in the following order: change in FPG from baseline at week 24; the proportion of participants with HbA1c less than 7.0% at week 24; change in LDL-C from baseline at week 24; change in non-HDL-C from baseline at week 24; change in GGT from baseline at week 24; and change in hs-CRP from baseline at week 24. Each end point was tested at a two-sided alpha level of 0.05 only if the preceding end point in the hierarchy achieved statistical significance. If significance was not achieved at any step, formal testing of subsequent end points was not performed, and only the point estimates and 95% CIs are provided. For secondary end points not considered in the gatekeeping strategy, only the point estimates and 95% CIs are provided, which have not been adjusted for multiplicity and should not be used in place of hypothesis testing.

Dichotomous end points were analyzed by using a nonresponder imputation method for missing data and analyzed by using a Cochran–Mantel–Haenszel model for the comparison between groups. Continuous variables were analyzed with a mixed-effects model for repeated measures similarly to the primary analysis. For the mixed meal tolerance test area under the concentration–time curve, a base-10 log transformation was performed, and the change from baseline was calculated and analyzed by using an analysis of covariance. A prespecified analysis of hs-CRP was performed with a mixed-effects model for repeated measures analysis similar to the secondary end points; the geometric mean ratio at week 24 was compared with baseline (natural logarithm scale) as the dependent variable and the baseline value (natural logarithm scale) as the covariate.

ROLE OF THE FUNDER/SPONSOR

HTD1801 was supplied by Shenzhen HighTide Biopharmaceutical. Authors from Shenzhen HighTide Biopharmaceuticals contributed to the design and conduct of the trial; collection, management, analysis, and interpretation of the data; and the preparation, review, and approval of

the manuscript. The first draft of the manuscript was written by a medical writer employed by the sponsor, under the direction of the authors; all the authors critically reviewed the manuscript, provided substantive input during drafting, contributed to revisions, and approved the final version. The decision to submit the manuscript for publication was made jointly by the authors, including those employed by the sponsor.

Results

PARTICIPANTS

Between November 2023 and July 2025, a total of 626 patients entered the run-in period, 549 patients were randomly assigned and treated (HTD1801, $n=365$; placebo, $n=184$) during the double-blind phase, and 498 patients completed the data collection protocol in the double-blind phase (Fig. S2). All 549 patients in the double-blind phase were included in the full analysis and safety sets. Although the 498 patients who completed the double-blind phase entered the open-label extension period, the open-label extension efficacy set comprised 497 patients because 1 patient did not receive the trial drug and did not complete any postbaseline efficacy assessments. A total of 480 patients completed the full 56-week period.

Demographic and baseline clinical characteristics are presented in Table 1. Baseline HbA1c was 8.6% and 8.5% in the HTD1801 and placebo groups, respectively. The mean ($\pm$ SD) daily dose of metformin was 1557.8 ± 242.4 mg/d in the HTD1801 group and 1509.8 ± 215.6 mg/d in the placebo group. Although all of the trial patients were Chinese, their baseline demographic and clinical characteristics are representative of the broader population with T2D (Table S1).

HBA1C VALUES

The baseline mean HbA1c was 8.6% in the HTD1801 group and 8.5% in the placebo group. At week 24, the mean HbA1c was 7.4% in the HTD1801 group and 7.8% in the placebo group. The mean change from baseline to week 24 was -1.2 percentage points (95% CI, -1.3 to -1.1) in the HTD1801 treatment group and -0.7 percentage point (95% CI, -0.8 to -0.6) in the placebo group (Fig. 1A). The adjusted LS mean difference between groups was -0.5 (95% CI, -0.7 to -0.4 ; $P<0.0001$) at week 24. The HbA1c reduction persisted in the HTD1801 group through 52 weeks (Fig. 1B). Sensitivity analyses were consistent with

Parameter	Placebo (n=184)	HTD1801 1000 mg BID (n=365)
Age — yr	57.6±9.1	56.1±10.1
Sex — no. (%)		
Male	97 (52.7)	199 (54.5)
Female	87 (47.3)	166 (45.5)
Ethnicity — no. (%)		
Han Chinese	173 (94.0)	355 (97.3)
Other	11 (6.0)	10 (2.7)
Weight, kg	70.3±12.0	69.9±11.0
BMI	26.2±3.2	25.9±3.1
Diabetes duration — yr	6.6±4.9†	6.0±4.7‡
Daily dose of metformin — mg/d	1509.8±215.6	1557.8±242.4
HbA1c — %	8.5±0.7	8.6±0.8
Fasting plasma glucose — mg/dl	163.6±32.9	164.7±35.9
2-hour postprandial plasma glucose — mg/dl	277.0±55.9	274.8±56.6
Fasting insulin — µU/ml	10.9±7.3§	11.6±9.0¶
Total cholesterol — mg/dl	174.2±36.7	180.1±37.6
HDL-C — mg/dl	45.6±9.8	45.5±10.7
Non-HDL-C — mg/dl	128.6±35.4	134.6±36.8
LDL-C — mg/dl	103.2±33.7	108.8±32.2
Triglycerides — mg/dl	153.5±87.4	159.2±102.0
Total bilirubin — mg/dl	0.8±0.3	0.8±0.3
ALT — U/l	20.7±11.8	21.3±14.5
AST — U/l	19.4±6.8	19.6±9.0**
GGT — U/l	27.8±16.7	30.9±28.1
hs-CRP — mg/l, median (Q1, Q3)	1.0 (0.6, 2.3)	1.0 (0.5, 2.4)

* Plus-minus values are means±standard deviation, unless otherwise noted; n values represent the total number of patients in each group. Full analysis set, N=549. Ethnicity was self-reported. BMI denotes body mass index (the weight in kilograms divided by the square of the height in meters). The upper limit of normal for the following measures are: non-HDL cholesterol (non-HDL-C), 158.3 mg/dl; LDL cholesterol (LDL-C), 130.7 mg/dl; total cholesterol, 200 mg/dl; and triglycerides, 150.4 mg/dl. SI conversion factors are as follows: to convert cholesterol to mmol/l, multiply values by 0.0259; glucose to mmol/l, multiply values by 0.0555; insulin to pmol/l, multiply values by 6.945; total bilirubin to mmol/l, multiply values by 17.104; and triglycerides to mmol/l, multiply values by 0.0113. ALT indicates alanine aminotransferase; AST, aspartate aminotransferase; BID, twice daily; GGT, gamma glutamyl-transferase; HbA1c, glycated hemoglobin; and hs-CRP, high sensitivity C reactive protein.

† n=180.

‡ n=364.

§ n=183.

¶ n=361.

|| n=364.

** n=362.

the primary analysis (Table S2). Subgroup analyses of the primary end point are presented in Figure S3.

KEY SECONDARY END POINTS

At week 24, the LS mean change in FPG was −24.6 mg/dl in the HTD1801 group and −12.2 mg/dl in placebo group, with a between-group difference of −12.4 mg/dl (95% CI, −17.3 to −7.6; $P<0.0001$) (Fig. 2A). Furthermore, at 24 weeks, a significantly greater proportion of patients treated with HTD1801 compared with placebo achieved target HbA1c levels less than 7% (n=121 [33%] vs. n=20 [11%];

relative risk, 4.1; 95% CI, 2.4 to 6.8; $P<0.0001$) (Fig. 2B). As shown in Figures 2C–2E, the HTD1801 group had significantly greater reductions in LDL-C (−13.6 vs. 0.2 mg/dl; between-group LS mean difference, −13.8 mg/dl; 95% CI, −18.7 to −8.9; $P<0.0001$), non-HDL-C (−11.9 vs. 6.7 mg/dl; between-group LS mean difference, −18.6 mg/dl; 95% CI, −24.1 to −13.0; $P<0.0001$), and GGT (−4.4 vs. 0.6 U/l; between-group LS mean difference, −5.0 U/l; 95% CI, −7.9 to −2.1; $P<0.01$) at week 24. Changes in hs-CRP at week 24 were similar between treatment groups (−0.6 vs. −0.3 mg/l; between-group LS mean difference, −0.3 mg/l;

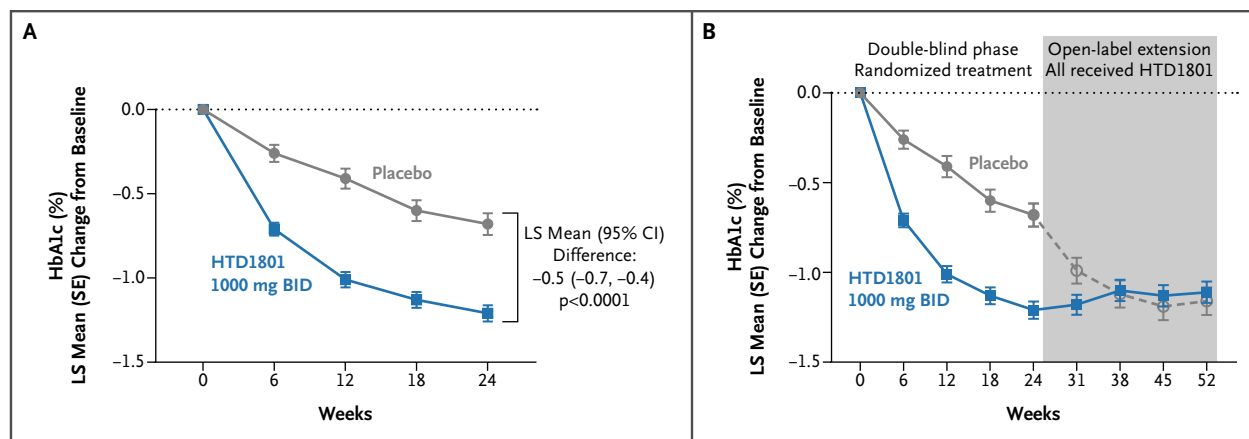


Figure 1. Change in HbA1c from Baseline.

Data are the adjusted least-squares (LS) means ($\pm$ SE) for the primary end point change in glycated hemoglobin (HbA1c) (%) from baseline to week 24 (Panel A) and the change in HbA1c (%) from baseline to the end of the open-label extension (week 52) (Panel B). All patients received HTD1801 1000 mg twice daily (BID) during the open-label extension. LS means are derived from a mixed-effects model for repeated measures, with the measured value or change from baseline as a dependent variable; the treatment group, measurement time point, interaction between treatment group and measurement time point, and high sensitivity C-reactive protein as independent variables; and the patient's baseline HbA1c value as a covariate. Vertical lines around point estimates represent standard errors. Double-blind phase full analysis set, n=549; open-label extension efficacy set, n=497.

95% CI, -0.8 to 0.2; $P=0.223$) (Fig. 2F). Results for other secondary efficacy measures are described in Sections 4 and 5 of the Supplementary Appendix, including Figs. S5, S6, S7, S8, and S9, as well as Tables S3 and S4.

OPEN-LABEL EXTENSION

At week 52, the change in HbA1c from baseline was -1.1 percentage points in the group treated with HTD1801 throughout the trial (i.e., double-blind and open-label extension periods) and -1.2 percentage points in the group who initially received placebo in the double-blind period and then switched to HTD1801 during the open-label extension period. In addition, switching to HTD1801, among patients in the placebo group of the double-blind period, was associated with reductions in other key secondary end points at 52 weeks, including FPG, LDL-C, non-HDL-C, GGT, and hs-CRP (Fig. 3). Upon switching to HTD1801, 41% of patients in the placebo group achieved target HbA1c levels less than 7%.

SAFETY

Adverse events in the double-blind phase are presented in Table 2. Adverse events occurred in 277 (75.9%) and 127 (69.0%) patients in the HTD1801 and placebo groups, respectively; the majority of these events were mild or

moderate in severity. One patient in the placebo group died of a craniocerebral injury. Serious adverse events occurred in 13 (3.6%) and 11 (6.0%) patients in the HTD1801 and placebo groups. Three patients receiving HTD1801 experienced a serious adverse event of diarrhea. Two patients in the placebo group and 7 patients in the HTD1801 group discontinued treatment due to adverse events.

The most common adverse event was diarrhea, which occurred in 87 (23.8%) patients in the HTD1801 group and 2 (1.1%) patients in the placebo group. The majority of the diarrhea events were graded as mild or moderate in severity. Three patients in the HTD1801 group experienced a severe case of diarrhea. The median duration of diarrhea was 6 days in both treatment groups. A total of 13 (3.6%) and 9 (4.9%) patients in the HTD1801 and placebo groups, respectively, required rescue therapy. An adverse event of hypoglycemia occurred in five patients in the HTD1801 group (mild, n=4; moderate, n=1) and 1 patient in the placebo group (mild, n=1). No patients in either group experienced severe hypoglycemia.

The adverse events through the entire 56-week follow-up period (open-label extension period [52 weeks] plus the final safety follow-up period [4 weeks]) are presented in Table S5. Similar to the double-blind period, diarrhea was the most common adverse event. However, among patients

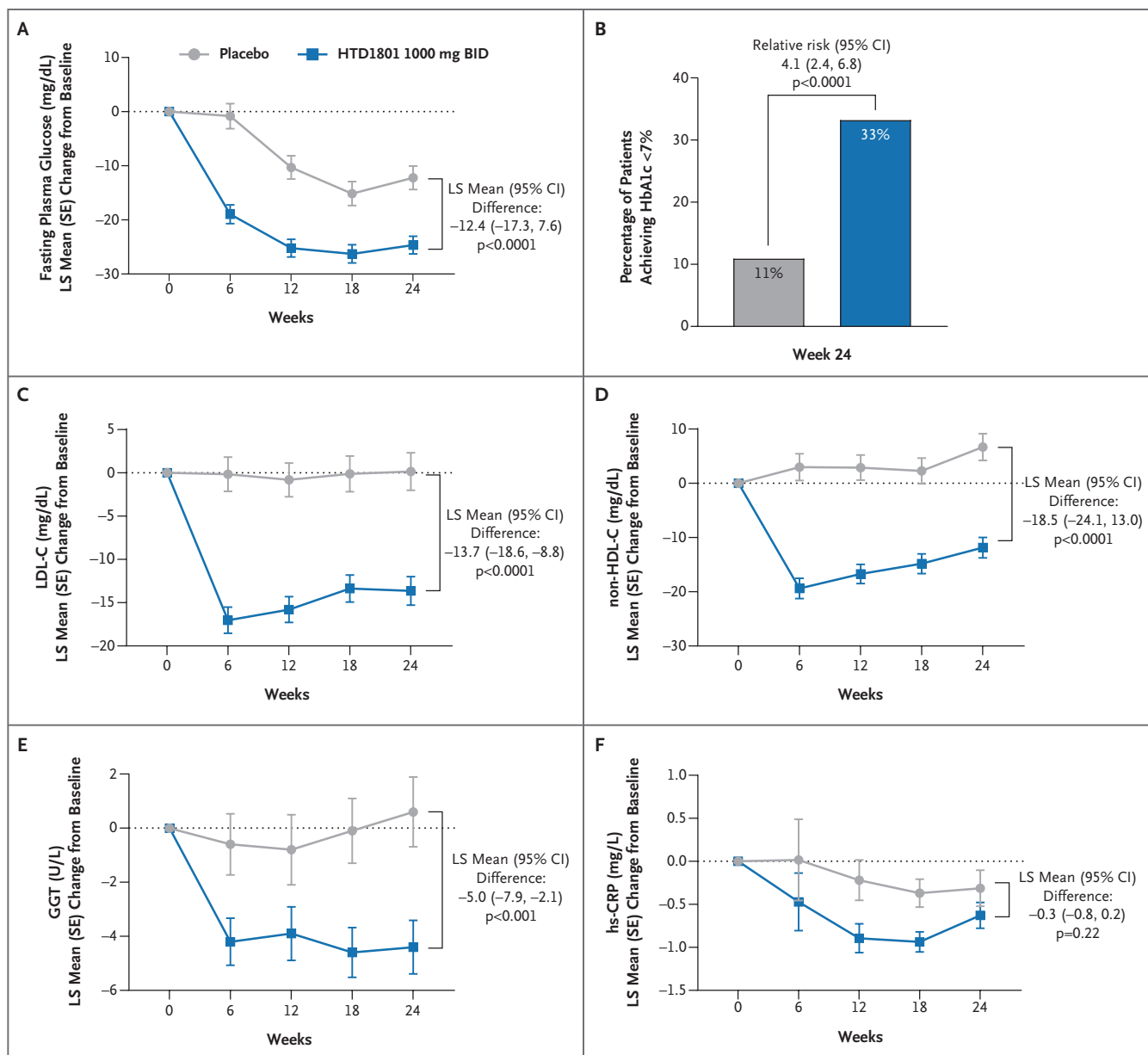


Figure 2. Key Secondary Efficacy End Points During Double-Blind Trial Period.

For key secondary end points, a type I error was controlled for by using prespecified sequential gatekeeping in the following order: change in fasting plasma glucose (Panel A); the proportion of participants with glycated hemoglobin (HbA1c) values less than 7.0% (Panel B); change in LDL cholesterol (LDL-C) (Panel C); change in non-HDL cholesterol (non-HDL-C) (Panel D); change in gamma glutamyl-transferase (GGT) (Panel E); and change in high sensitivity C-reactive protein (hs-CRP) (Panel F). Each end point was tested at a two-sided alpha level of 0.05 only if the preceding end point in the hierarchy achieved statistical significance. Least-squares (LS) means are derived from a mixed-effects model for repeated measures, with the measured value or change from baseline as a dependent variable, the randomization stratification factors as independent variables, and the patient's baseline HbA1c value as a covariate. For the percentage of patients achieving an HbA1c value less than 7%, a Cochran-Mantel-Haenszel model was used for comparison between groups, and the randomization stratification factors in the model were hs-CRP (≤ 3 mg/l or > 3 mg/l). The upper limit of normal for non-HDL-C is 158.3 mg/dl and for LDL-C it is 130.7 mg/dl. SI conversion factors were as follows: to convert glucose to mmol/l, multiply values by 0.0555; and to convert cholesterol to mmol/l, multiply values by 0.0259. Vertical lines around point estimates indicate standard errors. Double-blind phase full analysis set, N=549. BID indicates twice daily.

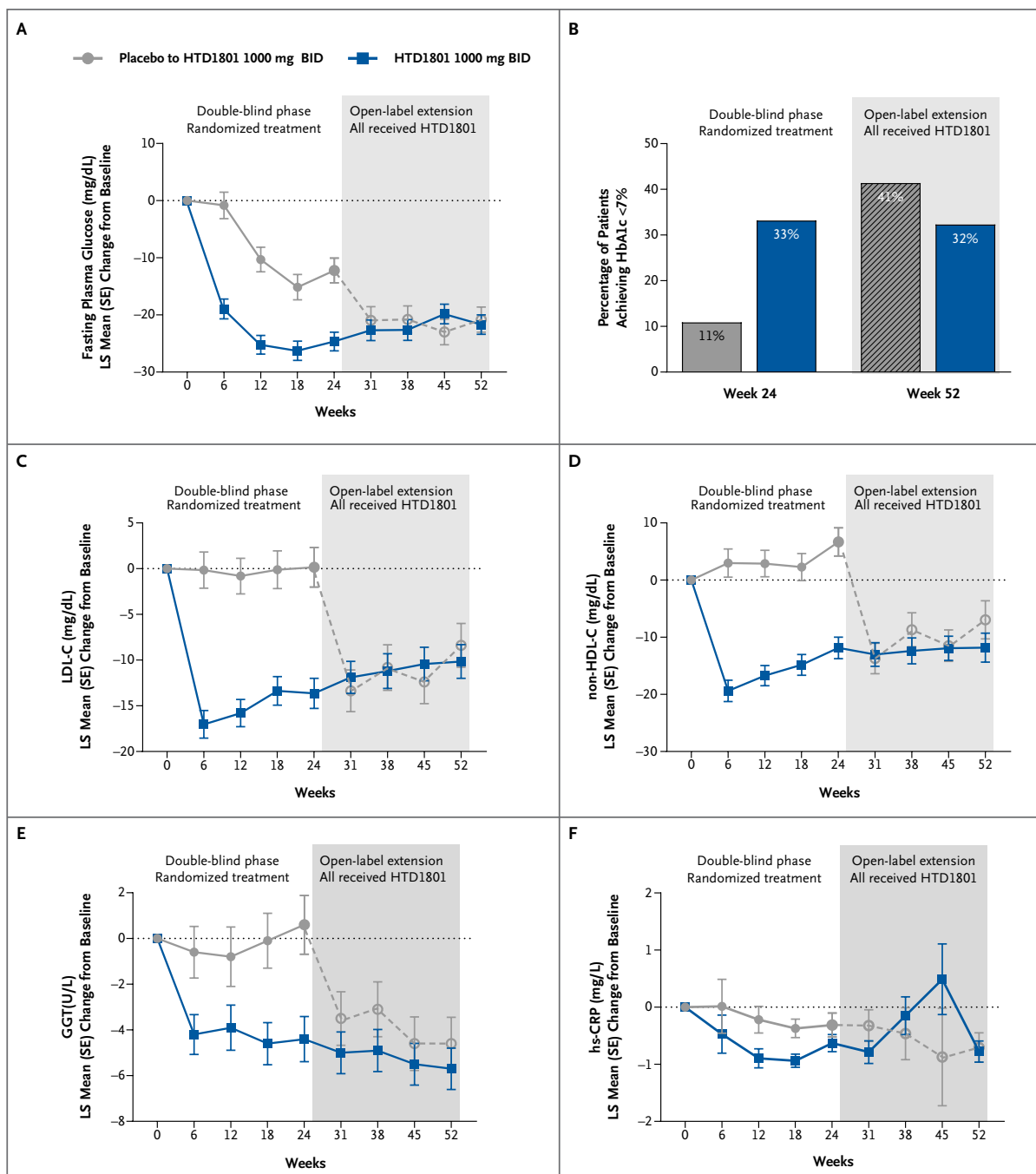


Figure 3. Key Secondary End Points — Double-Blind and Open-Label Extension Phases.

Data are the adjusted least-squares (LS) means ($\pm$ SE) for change in fasting plasma glucose (Panel A); The proportion of participants with glycated hemoglobin (HbA1c) values less than 7.0% (Panel B); change in LDL cholesterol (LDL-C) (Panel C); change in non-HDL cholesterol (non-HDL-C) (Panel D); change in gamma glutamyl-transferase (GGT) (Panel E); and change in high sensitivity C-reactive protein (hs-CRP) (Panel F). Least-squares (LS) means are derived from a mixed-effects model for repeated measures, with the measured value or change from baseline as a dependent variable, the randomization stratification factors as independent variables, and the patient's baseline HbA1c value as a covariate. For the percentage of patients achieving an HbA1c value less than 7%, a Cochran–Mantel–Haenszel model was used for comparison between groups, and the randomization stratification factors in the model were hs-CRP (≤ 3 mg/l or > 3 mg/l). The upper limit of normal for non-HDL-C is 158.3 mg/dl and for LDL-C it is 130.7 mg/dl. SI conversion factors were as follows: to convert glucose to mmol/l, multiply values by 0.0555; and to convert cholesterol to mmol/l, multiply values by 0.0259. Vertical lines around point estimates indicate standard errors. Double-blind phase full analysis set, N=549. Extended Treatment Period Efficacy Set (N=497). BID indicates twice daily.

who continued to receive HTD1801 during the open-label extension, the incidence of diarrhea appeared to decline over time (Fig. S4).

Discussion

In this phase 3, randomized placebo-controlled trial, the primary end point, reduction in HbA1c over 24 weeks, was significantly greater among patients with T2D taking metformin who were randomly assigned to receive HTD1801 versus placebo. In addition, key secondary end points capturing glycemic, cardiometabolic, and inflammatory markers were also significantly improved in the HTD1801 versus placebo groups at 24 weeks, with the exception of hs-CRP.

Improvements in the primary and secondary efficacy end points persisted through 52 weeks. In the double-blind phase, serious adverse events were uncommon in both groups. Diarrhea was the most common adverse event and affected a greater proportion of patients randomly assigned to receive HTD1801; the majority of such cases were mild to moderate in severity.

HTD1801 has produced clinically meaningful reductions in HbA1c (i.e., reductions of approximately 1.2–1.3 percentage points) across clinical settings, whether administered in combination with metformin (as shown in the current trial [SYMPHONY-2]) or as monotherapy (as shown in SYMPHONY-1).¹² In the current trial, HTD1801 in combination with metformin resulted in significantly greater HbA1c reductions compared with placebo, although a

Table 2. Adverse Events Over 24 Weeks in the Double-Blind Trial Period.*

Event	Placebo (n=184)	HTD1801 1000 mg BID (n=365)
Any adverse event	127 (69.0)	277 (75.9)
Grade 1: mild	85 (46.2)	168 (46.0)
Grade 2: moderate	27 (14.7)	91 (24.9)
Grade 3 or higher: severe	15 (8.2)	18 (4.9)
Any serious adverse event	11 (6.0)	13 (3.6)
Adverse event leading to trial discontinuation	2 (1.1)	7 (1.9)
Diarrhea	0	3 (0.8)
Abdominal distension	0	1 (0.3)
Mediastinal neoplasm benign	0	1 (0.3)
Gastritis chronic	0	1 (0.3)
Myocardial infarction	0	1 (0.3)
Prostate cancer	1 (0.5)	0
Angina pectoris	1 (0.5)	0
Major adverse cardiovascular event†	8 (4.3)	7 (1.9)
Fatal adverse event	1 (0.5)	0
Head injury	1 (0.5)	0
Adverse events affecting ≥5% of patients in either group		
Diarrhea	2 (1.1)	87 (23.8)
Mild	2 (1.1)	53 (14.5)
Moderate	0	31 (8.5)
Severe	0	3 (0.8)
Nephrolithiasis	9 (4.9)	28 (7.7)
Hyperuricemia	14 (7.6)	30 (8.2)
Hyperlipidemia	13 (7.1)	22 (6.0)
Urinary tract infection	12 (6.5)	17 (4.7)

* Values are presented as number (%). Safety analysis set, N=549. The Medical Dictionary for Regulatory Activities version 28.0 was used for adverse event coding, and National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0 was used for grading severity.²⁰ BID indicates twice daily.

† Includes cardiovascular death (encompassing fatal myocardial infarction and fatal stroke), nonfatal myocardial infarction, nonfatal stroke, or hospitalization for unstable angina.

decrease was also observed in the placebo group. The placebo response may reflect improved adherence to lifestyle recommendations, background metformin therapy, and frequent glycemic monitoring mandated by the trial protocol. Consistent findings were observed in SYMPHONY-1, in which HbA1c levels also declined in the placebo group, but the magnitude of reduction was significantly greater with HTD1801.¹² In the monotherapy trial, patients treated with HTD1801 achieved an LS mean HbA1c reduction of −1.3 percentage points, supporting the notion that HTD1801 lowers glucose levels independent of concurrent metformin therapy.

Current clinical guidelines recommend an HbA1c target of less than 7.0%, with individualized goals based on patient characteristics.^{21,22} Despite a relatively high baseline HbA1c (approximately 8.6%) value with stable metformin therapy, approximately one-third of patients in this trial treated with HTD1801 achieved an HbA1c level less than 7.0% within 6 months, compared with 11% of patients treated with metformin alone. Similarly, in patients who switched from placebo to HTD1801 in the open-label extension, 41% of patients achieved an HbA1c level less than 7.0% at 52 weeks. After a mixed meal tolerance test at week 24, HTD1801 was associated with reduced postprandial glucose and insulin levels (Fig S5 of the Supplementary Appendix). These findings, together with 24-week improvements in FPG and HOMA-IR, suggest that HTD1801 enhanced insulin sensitivity and glucose utilization, thereby reducing insulin secretion and driving the hypoglycemic effect.

Patients with T2D often experience multiple comorbidities beyond hyperglycemia that can increase cardiovascular and mortality risk.^{23,24} HTD1801 in combination with metformin showed a significant reduction by 24 weeks in LDL-C and non-HDL-C, which are well-established biomarkers for cardiovascular disease.²² In addition, despite a mean baseline weight of 70.0 kg, a reduction in weight was observed with HTD1801 (Table S4).

In this trial, HTD1801 in combination with metformin did not exhibit additional risk of severe hypoglycemia. The most common adverse events with HTD1801 were mild to moderate diarrhea (23.8%), most of which occurred at HTD1801 treatment initiation (Fig. S4). In the phase 3 T2D trial evaluating HTD1801 as monotherapy (SYMPHONY-1), the proportion of patients experiencing diarrhea was less than 10%, which may suggest that diarrhea is more common with combination therapy.¹² Notably, in patients continuing HTD1801 in the open-label extension period, the proportion of patients experience diarrhea was reduced

to approximately 4%, suggesting potentially improved tolerability with continued treatment.

This trial had several limitations. All patients in this trial were Chinese; however, their demographic and cardiometabolic characteristics were representative of adults with T2D receiving metformin in non-Chinese populations. In addition, the trial was not powered or specifically designed to evaluate major adverse cardiovascular events.

The current trial showed that treatment with HTD1801 in combination with metformin provides additional glycemic benefit to patients with T2D who were unable to achieve glycemic control with metformin alone. In addition, significant improvements in lipids and markers of inflammation support additional metabolic benefits of HTD1801. These findings are consistent with the known antidiabetic and metabolic effects of berberine, which have been shown to improve glycemic control, lipid metabolism, and inflammatory pathways through complementary mechanisms, including inhibition of gluconeogenesis, enhancement of insulin secretion, and anti-inflammatory actions.²⁵

In patients with T2D inadequately controlled with metformin, glucagon-like peptide-1-receptor agonists, sodium-glucose cotransporter-2 inhibitors, dipeptidyl peptidase-4 inhibitors, and insulin are all potential considerations. This trial found that among these patients with T2D, HTD1801 is an additional agent, that compared with placebo, significantly reduced HbA1c values and improved key metabolic markers at 24 weeks.

Disclosures

Author disclosures and other supplementary materials are available at evidence.nejm.org.

Topline data from the double-blind phase of this trial were presented at the 61st European Association for the Study of Diabetes (EASD) Annual Meeting; September 15 to 29, 2025; Vienna, Austria.

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